

What if we picked a different working fluid such as water.

For water: $\rho_w = 1000 \text{ kg/m}^3$
 $k_w = 0.6 \text{ W/m}\cdot\text{K}$
 $Pr = 7$
 $\nu_w = 0.658 \times 10^{-6} \text{ m}^2/\text{s}$

$$Re_L = \frac{V_\infty L}{\nu_w} = \frac{(2 \text{ m/s})(5 \text{ m})}{(0.658 \times 10^{-6} \text{ m}^2/\text{s})} = 1.52 \times 10^7 \text{ (Turbulent!)}$$

$$\overline{C_f} = 0.074 Re_L^{-1/5} - 1742 Re_L^{-1} = 0.00259$$

$$F_D = \overline{C_f} \cdot A \cdot \frac{\rho_w V_\infty^2}{2} = (0.00259)(5 \text{ m}^2) \frac{(1000 \text{ kg/m}^3)(2 \text{ m/s})^2}{2}$$

$$\boxed{F_D = 25.9 \text{ N}}$$

For heat transfer:

$$\overline{Nu_L} = \frac{\overline{h} L}{k_w} = (0.037 Re_L^{4/5} - 871) Pr^{1/3} = 35593.8$$

$$\overline{h} = \frac{\overline{Nu_L} \cdot k_w}{L} = 4271.3 \text{ W/m}^2 \cdot \text{K}$$

$$\boxed{Q = \overline{h} A (T_w - T_\infty) = 0.854 \text{ MW} \quad !!!}$$

$$\boxed{\frac{F_{D, \text{oil}}}{F_{D, \text{w}}} = 2.2}$$

$$\boxed{\frac{Q_{\text{oil}}}{Q_w} = 0.013 \text{ or } 1.3\% \text{ only!}}$$

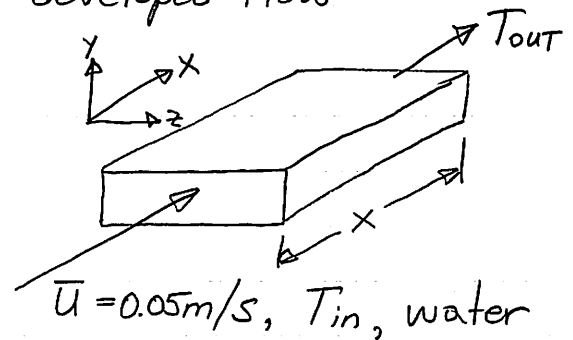
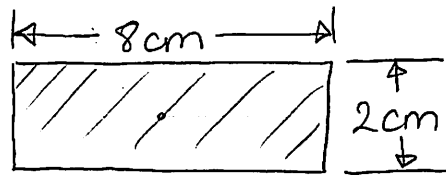
This simple analysis shows the importance of selecting the appropriate working fluid.

You may ask yourself, why would anybody use high Pr oils for thermal applications. Isn't water way better?

The answer is NO! Oils have 2 main advantages:

- 1) Temperature range (Typically -40°C to 300°C).
- 2) Low volatility. Water has a relatively high vapor pressure, meaning it will evaporate easily.
- 3) Corrosion resistance and enhanced lubrication.
- 4) Oil is a dielectric fluid (electrically insulating). Water is also a dielectric, however dissolved ions can readily make it into an electrolyte.

Example Heat exchange to a liquid flowing in a channel with $T_w = \text{constant}$ and fully developed flow:



Determine T_{out} as a function of x (length of channel)

First let's find the flow regime:

$$Re_D = \frac{\rho \bar{U} D}{\mu} \Rightarrow \text{But what } D \text{ do we use?}$$

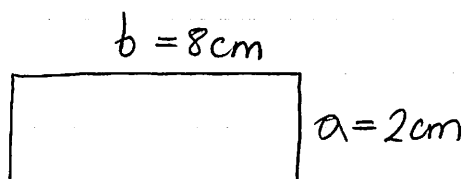
We need hydraulic diameter:

$$D_h = \frac{4A}{P} = \frac{4(8\text{cm})(2\text{cm})}{2(2\text{cm}) + 2(8\text{cm})} = \frac{64\text{cm}^2}{20\text{cm}} = 3.2\text{cm}$$

$$Re_{Dh} = \frac{\rho_w D_h \bar{U}}{\mu_w} = \frac{(1000 \text{ kg/m}^3)(0.032\text{m})(0.05 \text{ m/s})}{(0.001 \text{ Pa}\cdot\text{s})} = 1600$$

$$Re_{Dh} = 1600 < 2300 \Rightarrow \text{LAMINAR Flow}$$

So to solve for h , now we must look at table 8.1 on page of the class notes: (since non-circular cross section)

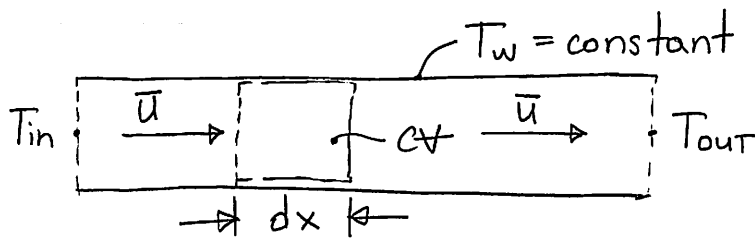


$$\frac{b}{a} = \frac{8}{2} = 4.0, \quad Nu_0 = \frac{h D_h}{k_f} = 4.44$$

↑
Uniform T_s

$$h = \frac{Nu_o \cdot k_f}{D_h} = \frac{(4.44)(0.6 \text{ W/m} \cdot \text{K})}{0.032 \text{ m}} = 83.25 \text{ W/m}^2 \cdot \text{K}$$

So now we can solve our problem. Let's look at a differential axial section of our channel:



Doing an energy ballance on our control volume:

$$\underbrace{\rho A \bar{u} c}_{\dot{m}} dT_f = \underbrace{h P dx}_{dA_{surface}} (T_w - T_f) ; \quad \begin{array}{l} c = \text{heat capacity} \\ A = \text{cross-sectional} \\ \text{area} \\ P = \text{perimeter} \end{array}$$

We want to convert this equation into a homogeneous ODE
Let's non-dimensionalize:

$$\text{Let: } T^* = \frac{T_f - T_w}{T_{in} - T_w} ; \quad x^* = \frac{x}{L} ; \quad L = \text{Length of channel}$$

$$dT^* = \frac{dT_f}{T_{in} - T_w} ; \quad dx^* = \frac{dx}{L}$$

Back substitute into our ODE:

$$\rho A \bar{u} c (\cancel{T_{in} - T_w}) dT^* = h P L dx^* (-T^*) (\cancel{T_{in} - T_w})$$

$$\rho A \bar{u} c dT^* = -h P L T^* dx^*$$